

Teaching Scheme and Syllabi of B.E. (Chemical Engineering)-MBA (2023-2028)

5th SEMESTER

S. No.	Course code	Courses	Contact hrs per week			Mid Term	End Term	Total Marks	Credits
			L	T	P				
1.	PCC 106	Chemical Engineering Thermodynamics	3	1	-	50	50	100	4
2.	PCC 108	Energy Technology	3	1	-	50	50	100	4
3.	PCC 110	Mass Transfer I	3	1	-	50	50	100	4
4.	PEC 101	Deptt. Elective I Numerical method in chem. Engg.	3	1	-	50	50	100	4
5.	MBA 107	Financial Accounting	4	-	-	50	50	100	4
6.	MBA 108	Business Analytics with R	3	1	-	50	50	100	4
7.	PEC 102	Department Elective-II Petroleum Processing Engg.	3	1	-	50	50	100	4
8.	PEC 151	Deptt. Elective Lab. I Numerical methods computation lab.	-	-	3	50	-	50	1.5
9.	PEC 152	Process Plant Design I	-	-	3	50	-	50	1.5
10.	PEC 153	Department Elective II Lab.	-	-	3	50	-	50	1.5
		Total	22	69		500	350	850	32.5

Credits – Chemical (24.5), MBA (8) Total hours: 37

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Paper Title: Departmental Elective (Theory)

Course Duration: 60 Lectures of one hour each.

Title	NUMERICAL METHODS IN CHEMICAL ENGINEERING		
THEORY			
Note for the Examiner	Question No. 1, which is compulsory, will cover the entire syllabus, having ten conceptual questions of one mark each or five questions of two marks each. Rest of the Questions (2 to 9) will be divided into FOUR Units having TWO questions each and candidate is required to attempt at least ONE question from each Unit. The duration of End Term exam will be 3 hrs.		
Course Objectives	To learn students: 1. Solve algebraic and transcendental equations, apply Least Square Curve Fitting Procedures to fit various curves and understand the concept of Finite differences. 2. Apply the concept of Finite differences to carry out Forward, Backward and Central Interpolation and Inverse Interpolation with equispaced and unequispaced data. Use suitable methods to perform numerical differentiation. 3. Use various methods to carry out numerical integration. Solve numerically ordinary differential equations of First order, higher order and Simultaneous differentialequations. 4. Solve linear system of equations by Direct and Iterative methods. Further, apply Finite Difference Approximation method to solve Partial differential equations.		
Course Outcomes	Upon successful completion of the course, the students will be able to: CO 1: Apply numerical methods for solving algebraic and transcendental equations, apply Least Square Curve Fitting Procedures to fit various curves and understand the concept of Finite differences. CO 2: Apply the concept of Finite Differences to carry out Forward, Backward and Central Interpolation and Inverse Interpolation with equispaced and unequispaced data. Also, apply suitable methods to perform numerical differentiation. CO 3: Use various methods to carry out numerical integration. Solve, numerically, ordinary differential equations of First order, higher order and Simultaneous differentialequations. CO 4: Solve linear system of equations by Direct and Iterative methods. Further, apply Finite Difference Approximation method to solve Partial		

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	differential equations.
Unit I	
Solution of Algebraic and Transcendental Equations: Bisection Method, Method of False Position, Iteration Method / Fixed Point Iteration Method, Newton-Raphson Method. 06 hrs.	
Curve Fitting: Least-Squares Curve Fitting Procedures for Fitting the Straight Line, the Second-degree Parabola and Fitting of other Curves. 04 hrs Finite Differences: Forward, Backward and Central Differences, Differences of a Polynomial, Factorial Notation. 05 hrs.	
Unit II	
Interpolation: Newton's Formulae for Forward and Backward Interpolation, Central Difference Interpolation Formulae: Stirling's Formula, Bessel's Formula, Interpolation with unevenly Spaced Points: Lagrange's Interpolation Formula, Divided Differences, Newton's Divided Difference Formula. 09 hrs. Inverse Interpolation: Lagrange's Method, Iterative Method. 03 hrs. Numerical Differentiation: Formulae for Derivatives, Maxima and Minima of a Tabulated Function. 03 hrs.	
Unit III	
Numerical Integration: Newton-Cotes Quadrature Formula: Trapezoidal Rule, Simpson's 1/3-Rule, Simpson's 3/8-Rule, Weddle's Rule. 04 hrs. Numerical Solution of Ordinary Differential Equation: Picard's Method, Euler's Method, Taylor's Series Expansion Method, Runge-Kutta Method, Predictor-Corrector Methods- Euler's modified method, Milne's Method. Simultaneous and Higher Order Equations: Runge-Kutta Method. 11 hrs.	
Unit IV	
Solution of Linear System of Equations: Gaussian Elimination Method, Gauss-Jordan Method, Jacobi Iteration Method, Gauss-Seidel Iteration Method, Condition of Convergence. 6 hrs. Numerical Solution of Partial Differential Equations: Classification of Second order equations, Finite-Difference Approximation to Partial Derivatives, Elliptic Equations- Solution of Laplace's Equation, Solution of Poisson's Equation, Parabolic Equations- Solution of one-dimensional heat equation (Schmidt method), Hyperbolic Equations- Solution of wave equation. 9 hrs.	
Books Recommended:	

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1.	Hildebrand, F.B.	:	Introduction to Numerical Analysis.
2.	Scarborough, J.B.	:	Numerical Mathematical Analysis, Oxford and ISH Pub. Co.
3.	Chopra, S.C., & Canale, R.P.	:	Numerical Methods for Engineers.
4.	Sastry, S. S.	:	Introductory Methods of Numerical Analysis, 4 th Edition, Prentice Hall.
5.	Grewal, B. S.	:	Numerical Methods in Engineering and Science.

PETROLEUM PROCESSING ENGINEERING (Theory)

THEORY		Time	3 Hours
Note for the Examiner	Question No. 1, which is compulsory, will cover the entire syllabus, having ten conceptual questions of one mark each or five questions of two marks each. Rest of the Questions (2 to 9) will be divided into FOUR Units having TWO questions each and candidate is required to attempt at least ONE question from each Unit. The duration of End Term exam will be 3 hrs.		

Course Objectives: The course aims at understanding the basic concepts of Petroleum Refining , refining operations and processes. Various aspects of refinery operations such as petroleum sources, technology and techniques, reaction mechanism, catalysts used and safety.

COURSE Outcomes

CO1 :Origin,exploration and drilling of petroleum crude and crude Pretreatment .

CO2 :learning the Evaluation methods of crude oil and properties and specifications of various petroleum products.

CO3 Understanding the separation processes and solvent extraction process.

CO4 Understanding the various conversion processes

Unit I

Introduction to petroleum industry, world petroleum resources, petroleum industry in India. Origin, exploration & drilling of petroleum crude. Crude pre-treatment: Refining and distillation of petroleum crude, composition and classification of petroleum crude

Unit II

Methods of evaluation: ASTM, TBP and EFV distillation. Properties and specifications of petroleum products such as LPG, gasoline, naphtha, kerosene, diesel, lubricating oils and waxes. Transportation of crude and products

Unit III

Separation Processes: Design and operation of topping and vacuum distillation units and tube still furnaces. Solvent extraction processes for lube oil base stock and for aromatics from naphtha and kerosene streams, solvent dewaxing.

Unit IV

Conversion Processes: Thermal cracking: visbreaking and coking processes, catalytic cracking, thermal reforming and catalytic reforming, alkylation, polymerization, isomerisation and hydroprocessing. Safety and pollution considerations in refineries.

Books Recommended: 1. Nelson, W.L. : Petroleum Refinery Engineering, 5th Edition, McGraw Hill, 1985. 2. Rao, B.K. : Modern Petroleum Refining Processes, 5th Edition, Oxford & IBH Publishing 2009. 3. Guthrie, V.B. : Petroleum Products Handbook, McGraw Hill, 1960. 4. Hobson, G.D., Pohl : Modern Petroleum Technology, 5th Edition, John Wiley, 1984.

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THIRD YEAR

5th SEMESTER

Title	CHEMICAL ENGINEERING THERMODYNAMICS			Credits	4
Code	PCC106			L T P	3 1 -
Max. Marks	End term- 50	Mid term- 50		Elective	N
Pre requisites					
THEORY				Time	3 Hours
Note for the Examiner	Question No. 1, which is compulsory, will cover the entire syllabus, having ten conceptual questions of one mark each or five questions of two marks each. Rest of the Questions (2 to 9) will be divided into FOUR Units having TWO questions each and candidate is required to attempt at least ONE question from each Unit. The duration of End Term exam will be 3 hrs.				
Course objectives	Chemical Engineering Thermodynamics is a core course which aims to link the concepts and laws of thermodynamics to the applications that require conceptual and quantitative knowledge of thermodynamic properties. It imparts understanding of phase equilibrium for designing of separation equipments. The course aims to use the techniques, skills, and modern engineering tools necessary for engineering practice				
Learning Outcomes	CO1: Understand the First and Second Laws of Thermodynamics apply it to open and closed systems, steady and unsteady state processes, isothermal and adiabatic processes and solve related engineering problems. CO2: Estimate the thermodynamic properties of pure substances, especially fluids. Knowledge of various PVT equations of state including Principle of corresponding states and heat capacities to evaluate thermodynamic properties of fluids. CO3: Explain the underlying principles of phase equilibrium and evaluate the thermodynamic properties in two-component and multi-component systems CO4: To develop and ability to envisage intermolecular potential and excess property behaviour of multi-component systems CO5: Impart ability to apply the concepts of phase equilibrium to vapour liquid equilibrium (VLE), separation processes and chemical reaction equilibrium				

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Unit I

Brief review of the terms: state functions, types of systems, internal energy, heat and work and reversible and irreversible processes. 1 hour

Use of Steam tables, First Law of Thermodynamics and its Engineering Applications i.e. constant volume processes, constant pressure processes, isothermal and adiabatic processes, pumps, turbines, compressors, nozzles, heat exchangers, pitot tube, venturimeter and orifice meter. 8 hours

Throttling Processes, Joule-Thomson Coefficient, liquefaction of gases 2 hours

Thermochemistry includes a brief review of heat capacities and their measurement, standard heat of reaction, standard heat of formation, standard heat of combustion, flame temperature, H-x diagrams, heat of solution, partial, molar enthalpies, enthalpy for phase change etc. 2 hours

Unit II

Equation of state for real gases and their mixtures. Principle of corresponding states and generalized compressibility factor. 3 hours

Review of Second law of thermodynamics, entropy concept, Entropy and lost work calculations. Microscopic interpretation of entropy. Third Law of thermodynamics and its applications. 3 hours

Free energy functions and their significance in phase and chemical equilibria, Clapeyron's equation and some important correlations for estimating vapor pressures. Estimation of thermodynamic properties by using graphs and tables. 4 hours

Unit III

Phase Equilibria: Partial molar properties, partial molar Gibbs free energy, Chemical potential and its dependence on temperature and pressure Ideal solutions (Lewis-Randel Rule). 6 hours

Fugacity and its calculations. Dependence of fugacity of temperatures and pressure 4 hours

Unit IV

Solution behaviour of real liquids and solids. Activity and activity coefficients. Variation of activity co-efficient with temperature and composition. Activity coefficients of electrolytes standard states. Properties of mixing. Residual and Excess Properties 4 hours

Gibbs-Duhem equation and its application to vapour-liquid equilibria. 2 hours

Chemical Equilibria:

Equilibrium constant in terms of measurable properties variations of equilibrium constant with temperature and pressure. 4 hours

Adiabatic reactions, Gibbs phase rule, equilibria in homogeneous reactions. 2 hours

Books Recommended:

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| 1. | Smith, J.M., Van Ness, H.C. and Abbott, M.M. | : | Introduction to Chemical Engineering Thermodynamics, 7 th Edition, McGraw Hill Professional, 2005 |
| 2. | Elliott, J.R and Lira, C.T. | : | Introductory Chemical Engineering Thermodynamic, Prentice Hall PTR., 1999. |
| 3. | Rao, Y.V.C. | : | Chemical Engg. Thermodynamics, Orient Blackswan, 1997. |
| 4. | Dodge, B.F. | : | Chemical Engg. Thermodynamics, McGraw Hill, 1944, Original from the University of Michigan, 2007. |
| 5. | Narayanan, K.V. | : | A Textbook of Chemical Engineering Thermodynamics, PHI Learning Pvt. Ltd., 2004. |

Title	ENERGY TECHNOLOGY			Credits	4
Code	PCC 108			L T P	3 1 -
Max.Marks	End term 50	Mid term 50	Practical	Elective	N
Pre requisites	-				
THEORY					
Note for the Examiner	Question No. 1, which is compulsory, will cover the entire syllabus, having ten conceptual questions of one mark each or five questions of two marks each. Rest of the Questions (2 to 9) will be divided into FOUR Units having TWO questions each and candidate is required to attempt at least ONE question from each Unit. The duration of End Term exam will be 3 hrs.				

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Course Objectives	➤ To make students understand various conventional and non-conventional energy resources. ➤ To make students solve the problems of combustion. ➤ To make students understand the working of various types of furnaces.
Course Outcomes	Upon successful completion of the course, the students will: CO1: Have in-depth knowledge of conventional and non-conventional fuels. Have knowledge of solid fuels, their origin, analysis, cleaning methods and carbonization process. Synthetic fuels from coal. CO2: Knowledge of liquid fuels, their origin, refining & distillation of crude oil. Manufacturing processes of gaseous fuels and their utilization. CO3: Become confident in solving combustion problems. Be able to describe various furnaces, draught and furnace atmosphere. CO4: Have in-depth knowledge of various renewable sources of energy, their scope and technologies in use.
UNIT- I (CO1)	
Fuels: Types of conventional fuels, their merits and demerits. Non-conventional/renewable energy sources, their importance for sustainable development and environmental protection. 5 Hrs Solid Fuels: Origin of coal, proximate and ultimate analysis of coal, coal preparation and washing methods, safe storage of coal. Low and High temperature carbonization, products of carbonization, By product coke ovens. Synthetics fuels from coal –Bergius process and Fischer Tropsch process. 10 Hrs	
UNIT- II (CO2)	
Liquid fuels: Origin of petroleum, refining and distillation of crude oil, uses of petroleum products. 7 Hrs Gaseous fuels: Natural gas, manufacture of water gas and producer gas, gas cleaning methods. 8 Hrs	
UNIT- III (CO3)	
Principles of combustion: Combustion calculations, waste heat utilization. 7 Hrs. Furnaces: Classification of furnaces, draught, furnace atmosphere, Portland cement continuous rotary kiln, blast furnace, glass melting furnace 8 Hrs.	
UNIT- IV (CO4)	
Alternate sources of energy: ✓ Introduction to solar radiation and evaluation of radiation incident on a solar collector. ✓ Applications of solar thermal energy such as solar water heater, solar concentrators and solar thermal power generation. ✓ Types of solar photovoltaic systems and applications ✓ Biomass Energy ✓ Wind Energy ✓ Geothermal Energy ✓ Tidal and Wave Energy 15 Hrs.	
Recommended Books	
1. Gupta, O.P. : Elements of Fuels, Furnaces & Refractories, 5 th Edition, Khanna Publishers, 2007. 2. Rao, S. and Parulekar, B.B. : Energy Technology – Non-conventional, Renewable & Conventional, 3 rd Edition, Khanna Publishers, 2007. 3. Dayal, M. : Renewable Energy – Environment and Development, Konark Publishers Pvt. Ltd., 1989. 4. Sukhatme, S.P. : Solar Energy – Principles of Thermal Collection and Storage, 2 nd Edition, Tata McGraw – Hill Publishing Company Ltd., 2006. 5. Sharma, S.P. and Mohan, C. : Fuels and Combustion, Tata Mc-Graw Hill Publishing Company Ltd., 1984.	

Title	MASS TRANSFER – I			Credits	4
Code	PCC110			L T P	3 1 -
Max.Marks	End term 50	Mid term 50	Practical 0	Elective	N
Pre requisites	-				
Course Objectives	The main purpose of the course is to provide fundamental understanding of basic				

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	principles of mass transfer in gases and in liquids and their applications in various mass transfer systems used in process and allied industries.
Course Outcomes	CO1: Classify mass transfer operations, laws of mass transfer, evaluation of molecular diffusion in gases, liquids and solids CO2: Discuss /Mass transfer coefficient, determination of diffusion coefficient, interphase mass transfer and estimation of number of stages CO3: Construction and working of gas-liquid contacting equipments. Analysis of drying and discuss the working of different types of dryers. CO4: Evaluation of humidification operations, psychometric chart, adiabatic saturation temperatures, wet bulb temperature, Types of cooling tower
THEORY	
Note for the Examiner	Question No. 1, which is compulsory, will cover the entire syllabus, having ten conceptual questions of one mark each or five questions of two marks each. Rest of the Questions (2 to 9) will be divided into FOUR Units having TWO questions each and candidate is required to attempt at least ONE question from each Unit. The duration of End Term exam will be 3 hrs.
Unit I Mass transfer operations, classification of mass transfer operations, choice of separation methods, methods of conducting mass transfer operations, design principles. Introduction to mass transfer and diffusion, molecular diffusion in gases and liquids, diffusion coefficients for gases and liquids, diffusion in solids, types of solid diffusion. Unit II Mass transfer coefficients, types of mass transfer coefficients, mass transfer coefficients in laminar flow, theories of mass transfer. Interphase mass transfer, concept of overall mass transfer coefficient. Unit III Working principle, construction and industrial applications of various gas liquid contacting equipments like sparged vessels, mechanically agitated vessels, tray towers, packed towers, spray chambers, Venturi scrubber. Principle of drying, batch drying, drying curve, constructional details and working of different dryers. Unit IV Humidification operations, psychometric chart, adiabatic saturation temperatures, wet bulb temperature, adiabatic operations, types of cooling towers.	
Recommended Books 1. Treybal, Robert E. : Mass Transfer Operations, 3rd Edition. McGraw-Hill, 1981. 2. Sherwood, T.K., Pifford, Robert L. and Wilke, Charles R. : Mass Transfer, McGraw-Hill. 3. Sharma, K.R. : Principles of Mass Transfer, Prentice Hall of India Pvt. Ltd., 2007. 4. McCabe, Warren L., Smith Juliam C. and Harriott, Peter : Unit Operations of Chemical Engg., 7th Edition, McGraw-Hill, 2005. 5. Coulson & Richardson : Chemical Engineering, Vol.I (6th Edition, 2009) and Vol. II. (5th Edition, 2006).	

Title	Financial Accounting				Credits	4
Code	MBA 107	Course Type	Core		L T P	4 0 -
Max.Marks	End term 50	Midterm 50		Practical --	Elective	N
Pre requisites	-					
Course Objective	The objective of this course is to enable students make financial statements, analyse them and evaluate the financial performance of firms.					
Course Outcomes	CO1: Understand the process of accounting as taken up in the organizations and Identify the main elements of financial statements CO2: Explain the differences between management, financial and cost accounting					

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	CO3: Analyse the financial statements and point out areas of concern in the financial statements. CO4: Identify the differences between traditional accounting and contemporary areas of accounting such as social and human resource accounting.	
THEORY		
Note for the Examiner	Question No. 1, which is compulsory, will cover the entire syllabus, having ten conceptual questions of one mark each or five questions of two marks each. Rest of the Questions (2 to 9) will be divided into FOUR Units having TWO questions each and candidate is required to attempt at least ONE question from each Unit. The duration of End Term exam will be 3 hrs.	
UNIT-I	Accounting: Accounting and its functions; Accounting as an information system; Basic Accounting Concepts and Accounting Conventions; Accounting Principles; Generally Accepted Accounting Policies (GAAP); Accounting Standards, Branches of Accounting: Financial Accounting; Cost Accounting; Management Accounting;	5 HOURS
UNIT-II	Accounting cycle: Accounting Equation; Accounting Structure; Types of Accounts. Rules regarding Journal Entries; Recording of Journal Entries; Ledger Posting; Cash book; Trial Balance; Preparation of Final Accounts; Trading Account; Profit & Loss Account; Balance Sheet; Treatment of Adjustments into trial balance.	10 HOURS
UNIT-III	Meaning of Management Accounting: Nature; Scope; Objectives; Functions of Management Accounting; Relationship between Financial and Management Accounting; Meaning of Financial Statement; Importance and Limitations of Financial Statement; Meaning and Objectives of Financial Statement Analysis; Limitation of Financial Analysis. Tools of financial analysis: Ratio analysis; Common size statements; Trend analysis; Fund flow and cash flow statement.	16 HOURS
UNIT-IV	Cost Accounting: Meaning, scope and classification of costs; Absorption costing; Marginal costing and break even analysis; Use of cost data in managerial decision making, Preparation of budgets and their control New branches of accounting: Responsibility Accounting- Responsibility Centre; Price Level Accounting, Social Accounting, Human Resource Accounting	14 HOURS
Text books	1. Charles T. Horngren, Gary L. Sundem, John A. Elliott and Donna Philbrick: Introduction to Financial Accounting, Prentice Hall India. 2. Hanif Mukherjee: Financial Accounting, Tata McGraw Hill	
Reference books:	1. Charles T. Horngren, Jeff Schatzberg, Gary L. Sundem and William O. Ostratton: Introduction to Management Accounting, Pearson Education. 2. Hilton Ronald: Managerial Accounting, McGraw Hill Professional 3. Charles T. Horngren, Srikanth M. Datar and George Foster: Cost Accounting- A Managerial Emphasis, Prentice Hall India. 4. N.R.Swamy: Financial Accounting-A managerial Perspective, Prentice Hall India. 5. J.R. Williams, S.F. Haka, M.S. Bettner and R.F. Meigs: Financial and Managerial Accounting – The Basis for Business Decisions, Tata McGraw Hill.	
Course Assessment	Assessment will consist of the following components 1. Mid-Term Assessment:	

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Methods	One best of two minor tests (50% of Mid -term marks) Assignments (20% of Mid-term marks) Class Surprise Tests/ Quizzes/Presentations/Term paper (20% of Mid-term marks) Attendance. (10% of Mid-term marks) 2.End –Term Assessment: University Examination
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Title	Business Analytics with R				Credits	04
Code	MBA 108	Course Type	Core		L T P	4 0 0
Max.Marks	End term 50	Midterm 50	Practical --		Elective	N
Pre requisites	Business Statistics					
Course Objectives	The objective of this course is to familiarize the students with different aspects of business analytics and solve various business problems using R.					
Course Outcomes	CO1: Gain Knowledge of basic concept / fundamentals of business statistics and fundamentals of R CO2: To understand and apply various data types and structures in business problems. CO3: To perform data visualization and apply Hypothesis Testing concepts and inferential statistics CO4: To perform practical application by taking managerial decision and using the concepts such as linear regression, dimension reduction techniques etc.					
Note for the student	Mandatory for the student to carry laptop in the class					
Note for the Examiner	Question No. 1, which is compulsory, will cover the entire syllabus, having ten conceptual questions of one mark each or five questions of two marks each. Rest of the Questions (2 to 9) will be divided into FOUR Units having TWO questions each and candidate is required to attempt at least ONE question from each Unit. The duration of End Term exam will be 3 hrs.					
UNIT-I	Business Analytics Basics: Concept, evolution, need, Business analytics vs business analysis, Business intelligence vs Data Science, Data Analyst Vs Business Analyst, Importance of data in business analytics, Differences between data, information and knowledge, various stages of an organization in terms of data maturity, Options for organizations in the absence of good quality data. Analytical decision-making: Analytical decision-making process, characteristics of the analytical decision-making process. Breaking down a business problem into key questions that can be answered through analytics, Overview of Business analytics applications across different fields - Marketing Analytics, HR Analytics, Supply Chain Analytics, Retail Industry, Sales Analytics, Web & Social Media Analytics, etc. Fundamentals of R: R environment, Downloading and Installing R, Using command line in R, Help, File operations in R -Reading from and Writing to a file, Writing your first code in R, Importing data from spreadsheets, text files.					10 HOURS

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UNIT-II	Data types & Data Structures in R: Data types in R and its appropriate uses, Program Structure in R, Flow Control: For loop, If condition, While conditions and repeat loop, Debugging tools, Concatenation of Data, Combining Vars , cbind, rbind, Sapply, apply, tapply functions, Built-in functions in R like: seq(), cbind (), rbind(), merge(), knowledge on the various subsetting methods, summarize data by using functions like: str(), class(), length(), nrow(), ncol(), use of functions like head(), tail(), for inspecting data, summarize data in R. Introduction to Data Structure in R, Vectors, Lists, Scalars, Data Frames, Matrices, Arrays, Factors, Use of data structures in different conditions, Advantage of using a particular approach.	12 HOURS
UNIT-III	Data Visualization: Concept of Data Visualization, Popular Data Visualization tools, Exploratory Data Analysis (EDA), Data Cleaning, Data Inspection, uses of the functions like grepl(), grep(), sub(), summarize(), llist(), Using graphical functions in R for data visualization, Line Plots, Bar Plots, Bar Plots for Population, Pie chart, tableplot, histogram. Statistics with R: Computing basic statistics, Business Hypothesis Testing concepts, Basics of statistical modeling, Comparing means of two samples, Testing a correlation for significance, Testing a proportion, t test, z Test, F test, Basics of Analysis of variance (ANOVA), One way ANOVA, ANOVA with interaction effects, Two way ANOVA, Summarizing Data, Cross tabulation	11 HOURS
UNIT-IV	Linear Regression: Concept of Linear regression, Dependency of variables, Ordinary Least Sum of Squares Model, Multiple Linear Regression, Obtaining the Best fit line, Assumptions and Evaluation, Outliers and Influential Observations, Multi-collinearity, Logistic Regression Dimension Reduction Techniques – Concept of latent dimensions, need for dimension reduction, Principal Components Analysis, Factor Analysis.	12 HOURS
Text Books:	1. G C Beri – Business Statistics, 3rd ed, TATA McGrawHill 2. Chandrasekaran & Umapparvathi-Statistics for Managers, 1st edition, PHI Learning 3. Davis , Pecar – Business Statistics using Excel, Oxford 4. Ken Black – Business Statistics, 5th ed., Wiley India 5. Levin and Rubin – statistics for Management, 7th ed., Pearson	
Course Assessment Methods	Assessment will consist of the following components 1. Mid-Term Assessment: - One best of two minor tests (50% of Mid -term marks) - Assignments (20% of Mid-term marks) - Class Surprise Tests/ Quizzes/Presentations/Term paper (20% of Mid-term marks) - Attendance. (10% of Mid-term marks) 2.End –Term Assessment: University Examination	

Title	PROCESS PLANT DESIGN –I			Credits	
Code	PEC 152			L T P	- - 3
Max. Marks	End term	Mid term	Practical: 40	Elective	N
Pre requisites	-				
THEORY					
Note for the Examiner					
Course Objectives	The course objective is to give the knowledge to the students about design of various equipments pumps, fans , blowers, settling equipments, agitated vessels and Conveyor system for solids				
Course Outcomes	CO1: Ability to understand the design of piping & piping networks CO2: The students are able to handle the design of various equipments like Pumps, Fans				

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	& Blowers. CO3: Ability to understand the design of settling equipments and agitated vessels . CO4 : Ability to understand design of Conveyor system for solids. .
Practical 1. Design of piping & piping networks. 2. Selection, specification & power requirements of process pumps, fans and blowers. 3. Design of settling equipments like Dor thickeners, dust chambers, cyclone separators and centrifuges. 4. Design of agitated vessels using various types of impellers. 5. Design of Conveyor system for solids.	
Recommended Books 1. Ludwig, E.E. : Applied Process Design in Chemical in Petrochemical Plants, Gulf Publishing Company. 2. Perry, J.H. : Chemical Engineers Handbook, McGraw Hill. 3. Joshi, M.V. : Process Equipment Design, Macmillan Indian. 4. Peters, M.S. and Timmerhaus, K.D. Plant Design and Economics for Chemical Engineers McGraw Hill.	